PURPLE TEAM HOME LAB · TECHNICAL REPORT

Lab v3 — Network Detection and Host EDR Layers

A full Active Directory kill chain run against five telemetry sources, then measured: what each layer caught, what it missed, and which of the shipped detections could not have fired at all.

Domain MARVEL.LOCAL · 7 hosts · VMware NAT 192.168.182.0/24

Attack run 20 October 2026 · **Rule verification** 24 October 2026

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Repository github.com/slim92code/purple-team-lab

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7. Detection engineering — rules shipped, tuned and verified

8. Alert triage — seven case files

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1. Summary

v3 adds two detection layers to the v2 Sysmon and Splunk baseline: network detection with Zeek and Suricata, and host EDR with Wazuh. Five telemetry sources feed one SIEM. The same kill chain was re-run and each phase checked against every source.

The interesting part is not that the attack was detected. It is what the measurement afterwards showed about the detections themselves. Three of the rules in the stack could not have alerted on this attack, for three different reasons, and all three looked correct until they were tested.

What changed from v2

Area	v2	v3
Telemetry sources	Windows Security log, Sysmon	plus Zeek, Suricata, Wazuh
Kerberos encryption	RC4 (0x17) roast	AES-256 (0x12) roast
Initial access lands on	a Domain Admin, so no escalation path	`pparker`, a standard user, who then roasts a DA SPN
Defender exclusions	`C:\Temp` excluded for reproducible delivery	none; `Get-MpPreference` shows no exclusion paths
Splunk licence	Free, so searches run on demand	Enterprise, so alerts fire on schedule
Deliverable	attack narrative and detection cards	seven alert-triage case files

The four findings that matter

> **A canonical detection that cannot fire.** The RC4 Kerberoasting rule keys on `Ticket_Encryption_Type = 0x17`. Counted over 30 days in this lab: 1,649 service tickets, every one of them `0x12`, zero RC4. The rule is scheduled, it sits in the coverage table, and it has never had an input.

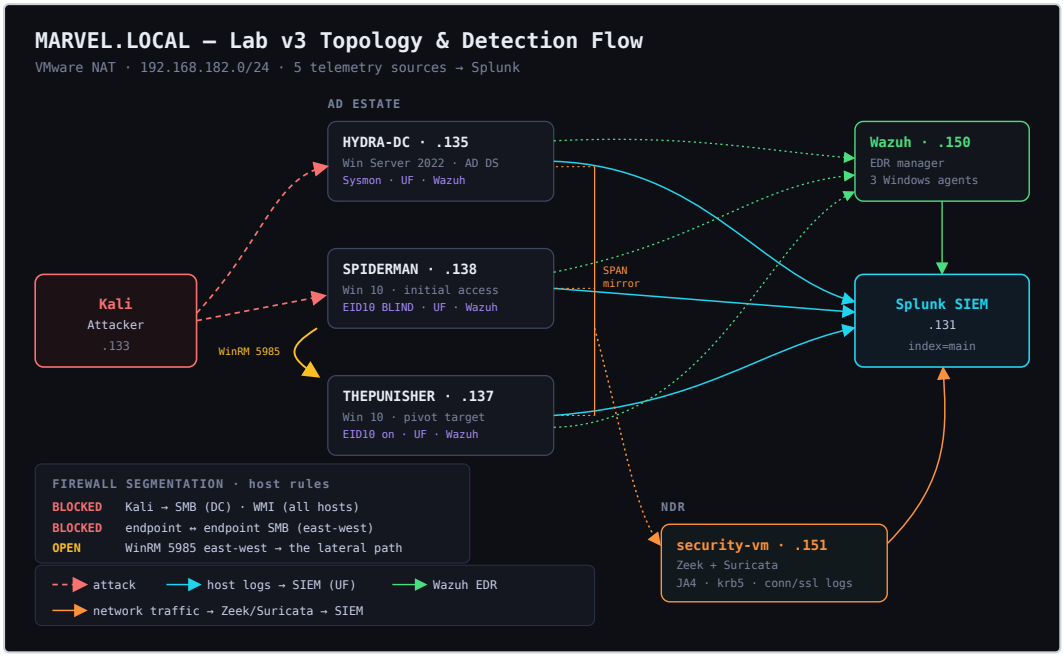
> **An EDR that stayed silent on a credential dump, for three separate reasons.** The other endpoint cannot produce the event at all, the EDR never receives Sysmon from two of three agents, and the built-in LSASS rule filters to the access mask Defender uses rather than the one a dump uses. A custom rule written to fix it bound to a rule group that does not exist, and the engine had been printing that warning on every start.

> **An alert that worked and still told the analyst nothing.** SYSTEM-level persistence with a PowerShell download cradle arrived at level 4, the same score the same rule gave a OneDrive updater. Scoring task creation without scoring task content discards the only part that separates the two.

> **A showcase detection querying an empty data model.** The behavioural C2 rule runs against `Network_Traffic.All_Traffic`, which holds zero events here. 444 Sysmon EID 3 events reach the index in seven days carrying no CIM fields, because the forwarder ships every Windows channel as a bare `WinEventLog` sourcetype and the Sysmon add-on keys its mappings on the per-channel sourcetype.

Each of these was found by measuring rather than by reading the rule. Each fix is verified by triggering the condition rather than by the absence of alerts. Section 7 and Appendix A carry the verification log.

2. Infrastructure



Attack paths in red, the WinRM pivot in amber, Wazuh agent channels in green, and the SPAN mirror feeding Zeek and Suricata. The firewall panel shows what the segmentation mesh blocks and what it leaves open.

Host	Address	Role
HYDRA-DC	`.135`	Windows Server 2022, AD DS, `MARVEL.LOCAL`
SPIDERMAN	`.138`	Windows 10 endpoint
THEPUNISHER	`.137`	Windows 10 endpoint
Splunk	`.131`	Splunk 10.2, Enterprise licence
Wazuh	`.150`	Wazuh 4.12 manager, 3 Windows agents
security-vm	`.151`	Zeek 9.0.0 and Suricata 6.0.4
Kali	`.133`	Attacker

Telemetry sources and what each one is good for

Source	Covers	Blind to
Windows Security log	authentication, process creation, task creation	an authentication attempt against an account that does not exist
Sysmon	process, network and handle-level detail on the host	whatever the per-host config does not include
Wazuh	host EDR, rule scoring on arriving events	channels the agent is not configured to read
Zeek	connection and protocol metadata, both outcomes of an exchange	anything that does not cross the mirrored link
Suricata	application-layer anomalies and signature matches	encrypted payload content

> **Agent count is not telemetry coverage.** Wazuh reports three healthy Windows agents. Across every archived alert file, only one of them has ever sent a Sysmon event. Section 6.4 carries the numbers.

3. Hardening baseline as built

The Windows estate is not a default install. A host-firewall segmentation mesh and SMB hardening were applied before the run, so the chain had to work through them.

Host	Blocks inbound, by explicit rule
HYDRA-DC	Kali to SMB, Kali to WMI
THEPUNISHER	SPIDERMAN to SMB, Kali to WMI
SPIDERMAN	THEPUNISHER to SMB, Kali to WMI

Also applied: SMBv1 removed on all three hosts, SMB signing required on the DC, account lockout at five attempts, and RemoteRegistry plus several discovery and legacy services disabled on the endpoints.

Where the protection actually comes from

SMB from the attacker to the two endpoints is blocked as well, but by the Windows inbound default rather than by a rule: their firewall profiles are `NotConfigured`, which means block. The DC's inbound default is `Allow`, so there the explicit rule is the only thing stopping it. One host is protected by a rule that someone wrote, two by a default nobody set.

Where it held and where it leaked

The mesh moved the attack off SMB and WMI. WinRM on 5985 was left open east-west, and that is the path the lateral movement took. Prevention narrowed the attack surface to a single predictable lane, and the detection layer was watching that lane. Narrowing is worth doing even when it does not block, provided something watches what is left open.

Residual gaps, deliberate, and the v4 before-state

Gap	Consequence
WinRM open east-west	the lateral-movement path; kept by design as the monitored pivot
SMB signing only on the DC	endpoints accept unsigned SMB
DC inbound default is `Allow`	only an explicit rule blocks the attacker there
No LSA Protection (`RunAsPPL`)	the LSASS dump succeeds, which is what section 5 detects
WDigest and LLMNR not hardened	credential-exposure and name-poisoning surface

4. Attack chain

The red-team side of the run. Each step produced the telemetry evaluated in section 5. Defender was enabled throughout, with no exclusion paths, which step 13 shows.

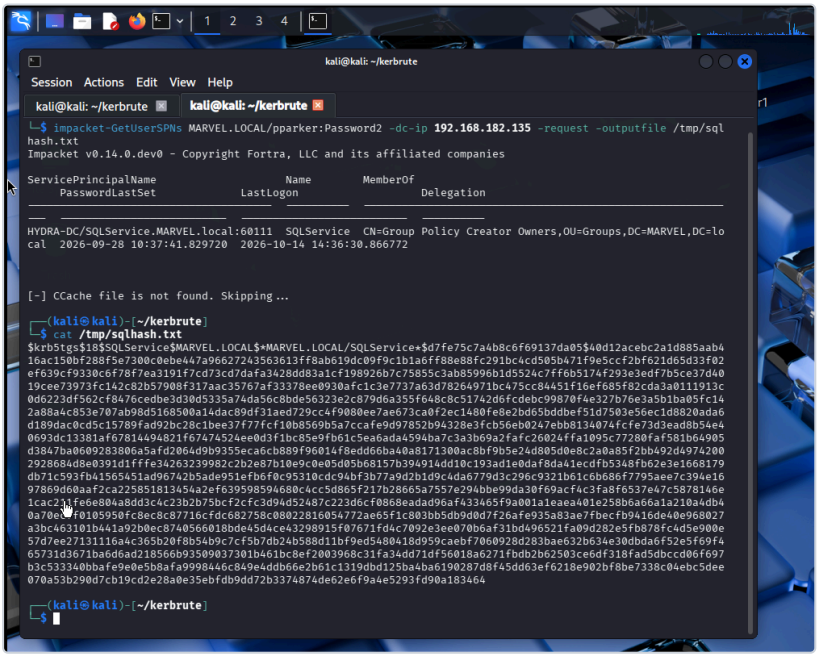
Lab credentials and a deliberately weak password are shown as captured. This is an isolated `MARVEL.LOCAL` lab.

Step 1 — Password spray (T1110.003)



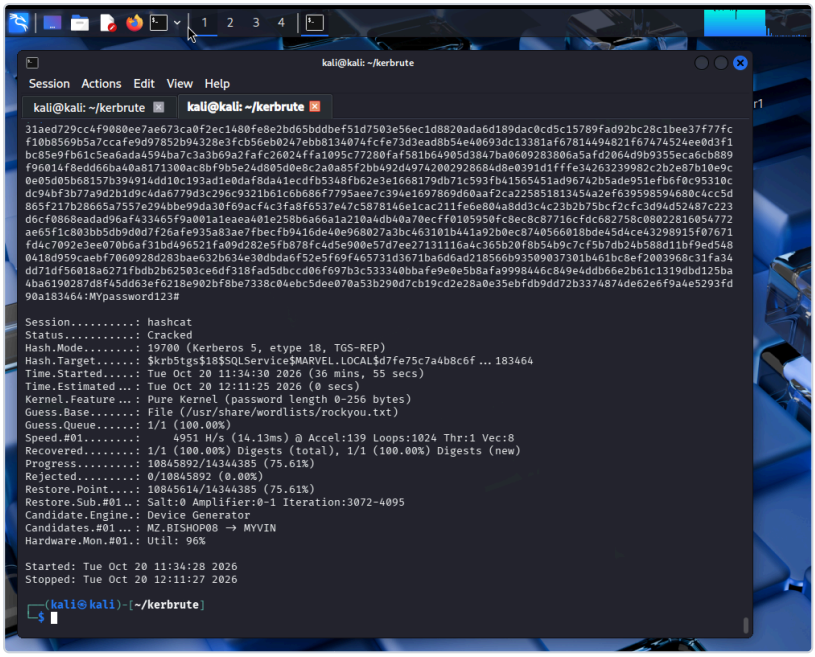
`kerbrute passwordspray` against the DC with one password across a user list. Seven logins tested, `pparker` valid. One attempt per account, which stays under the lockout threshold of five.

Step 2 — Kerberoasting, SPN ticket request (T1558.003)



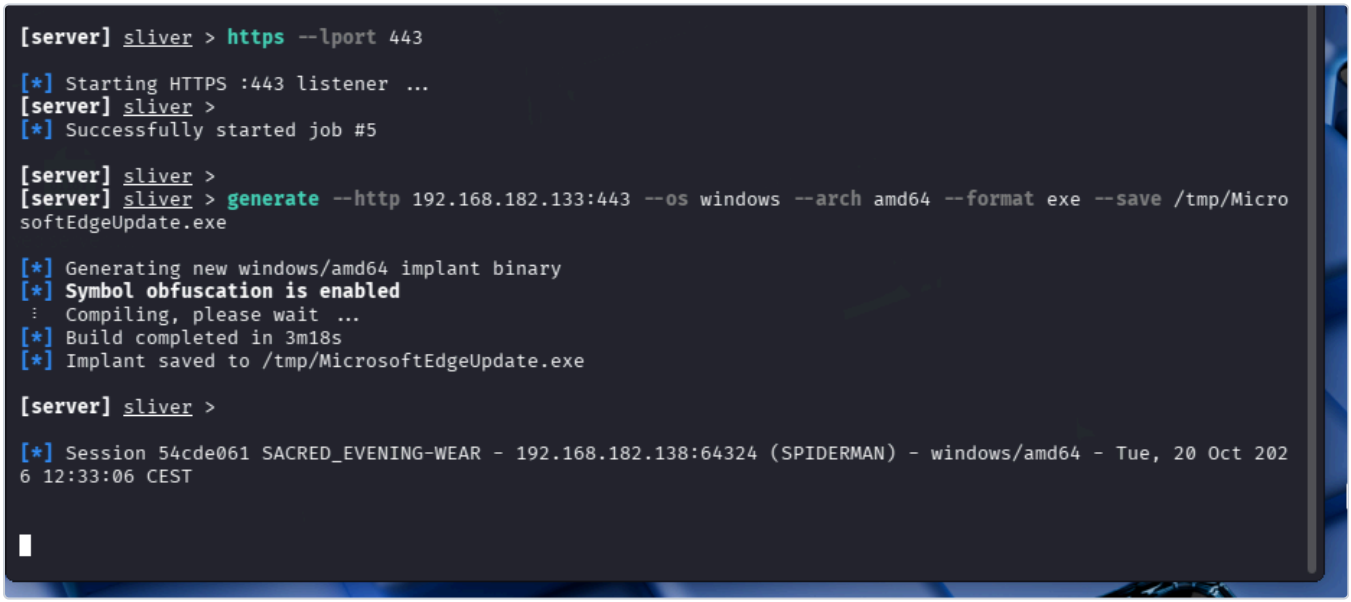
`impacket-GetUserSPNs` with the `pparker` credential requests the `SQLService` ticket and writes the `krb5tgs` hash.

Step 3 — Kerberoasting, offline crack (T1558.003)



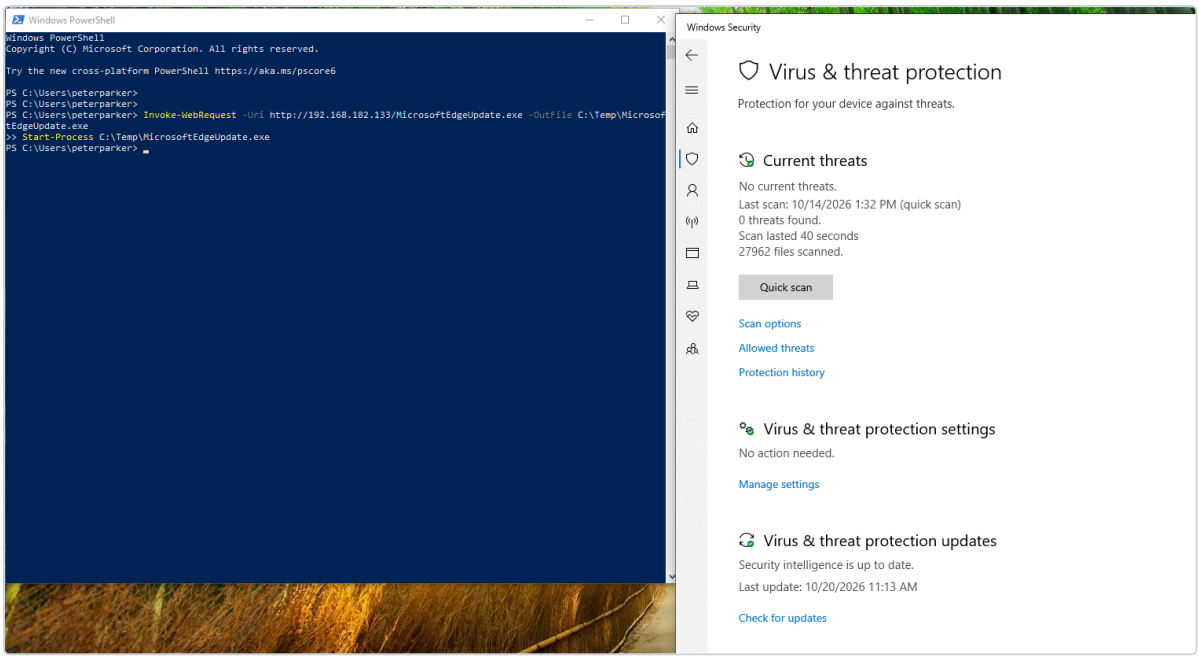
hashcat mode 19700, Kerberos 5 AES-256 TGS-REP, cracks the `SQLService` hash from a wordlist. The mode number is the point: this is an AES roast, not RC4.

Step 4 — C2 listener and implant (T1071.001)



Sliver HTTPS listener started and a Windows implant generated. The session from SPIDERMAN checks in.

Step 5 — Payload delivery (T1105)



The implant is downloaded to the endpoint and run. The Defender pane shows no detection, so what catches this is the detection layer rather than prevention.

Step 6 — Pass-the-Ticket (T1550.003)



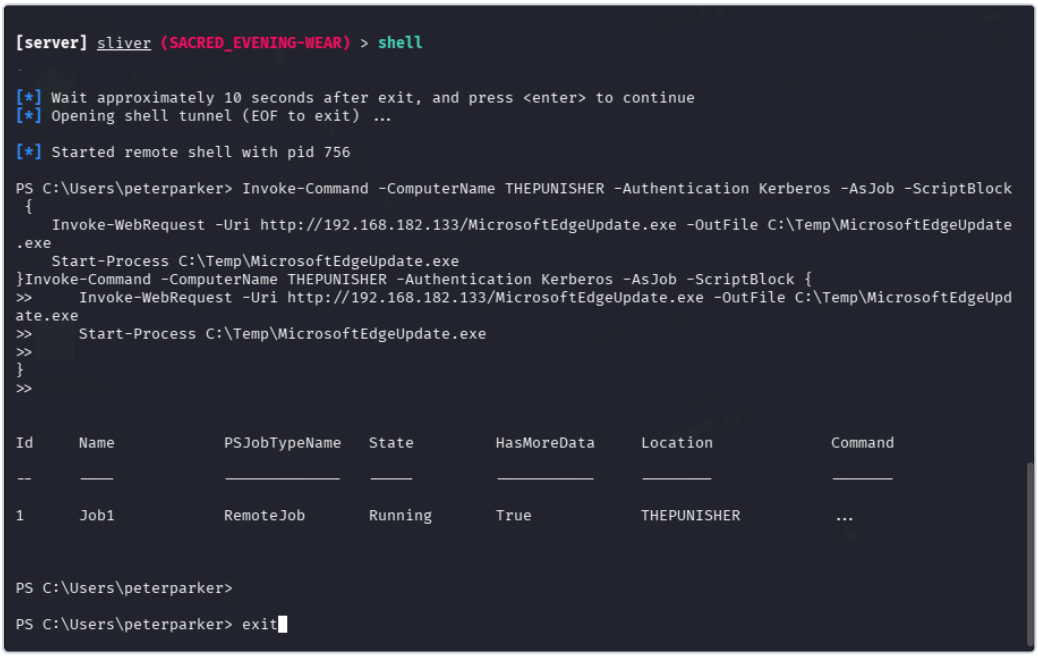
Rubeus run in memory through Sliver `execute-assembly`, requesting a TGT for `SQLService` and injecting it with `/ptt`.

Step 7 — Pass-the-Ticket, ticket imported (T1550.003)



The imported ticket with its flags and lifetime. The credential is now usable without the account password.

Step 8 — Lateral movement over WinRM (T1021.006)



`Invoke-Command -ComputerName THEPUNISHER -Authentication Kerberos` runs a remote job using the injected ticket. This is the port-5985 traffic in detection 05.

Step 9 — Pivot confirmed (T1021.006)

```
[server] sliver (SACRED_EVENING-WEAR) >

[*] Session 196725b2 SACRED_EVENING-WEAR - 192.168.182.137:53848 (THEPUNISHER) - windows/amd64 - Tue, 20 Oct 2
026 12:50:57 CEST


[server] sliver (SACRED_EVENING-WEAR) > sessions

 ID         Transport   Remote Address           Hostname      Username      Operating System   Health
==========  ==========  ======================  ============  ============  =================  ========
 54cde061   http(s)     192.168.182.138:64324   SPIDERMAN     peterparker   windows/amd64      [ALIVE]
 196725b2   http(s)     192.168.182.137:53848   THEPUNISHER   frankcastle   windows/amd64      [ALIVE]

[server] sliver (SACRED_EVENING-WEAR) > use 196725b2

[*] Active session SACRED_EVENING-WEAR (196725b2-242b-43a9-9544-f7a569db1fc4)

[server] sliver (SACRED_EVENING-WEAR) >
```

A second C2 session on THEPUNISHER alongside the SPIDERMAN session.

Step 10 — LSASS dump via SharpKatz (T1003.001)

```
[server] sliver (SACRED_EVENING-WEAR) > sessions

 ID         Transport   Remote Address           Hostname      Username      Operating System   Health
==========  ==========  ======================  ============  ============  =================  ========
 54cde061   http(s)     192.168.182.138:64324   SPIDERMAN     peterparker   windows/amd64      [ALIVE]
 196725b2   http(s)     192.168.182.137:53848   THEPUNISHER   frankcastle   windows/amd64      [ALIVE]

[server] sliver (SACRED_EVENING-WEAR) > use 196725b2

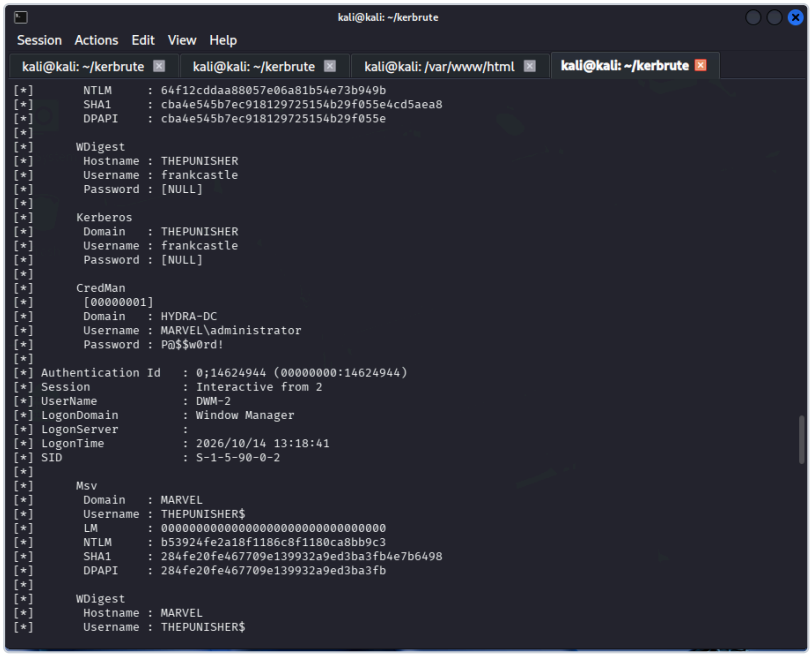
[*] Active session SACRED_EVENING-WEAR (196725b2-242b-43a9-9544-f7a569db1fc4)

[server] sliver (SACRED_EVENING-WEAR) > execute-assembly -t 120 /var/www/html/SharpKatz.exe --Command logonpas
swords

[*] Output:
[*]
[*]                    System Information
[*] ------------------------------------------------------------
[*] | Platform: Win32NT                                        |
[*] ------------------------------------------------------------
[*] | Major: 10           | Minor: 0            | Build: 19045 |
[*] ------------------------------------------------------------
[*] | Version: Microsoft Windows NT 10.0.19045.0               |
[*] ------------------------------------------------------------
[*]
[*] Authentication Id   : 0;32202442 (00000000:32202442)
[*] Session             : Network from 0
[*] UserName            : SQLService
[*] LogonDomain         : MARVEL
[*] LogonServer         :
[*] LogonTime           : 2026/10/20 12:42:51
[*] SID                 : S-1-5-21-1422081359-1692309262-2219121369-1104
[*]
[*]      Kerberos
[*]       Domain   : MARVEL.LOCAL
[*]       Username : SQLService
[*]       Password : [NULL]
[*]
```

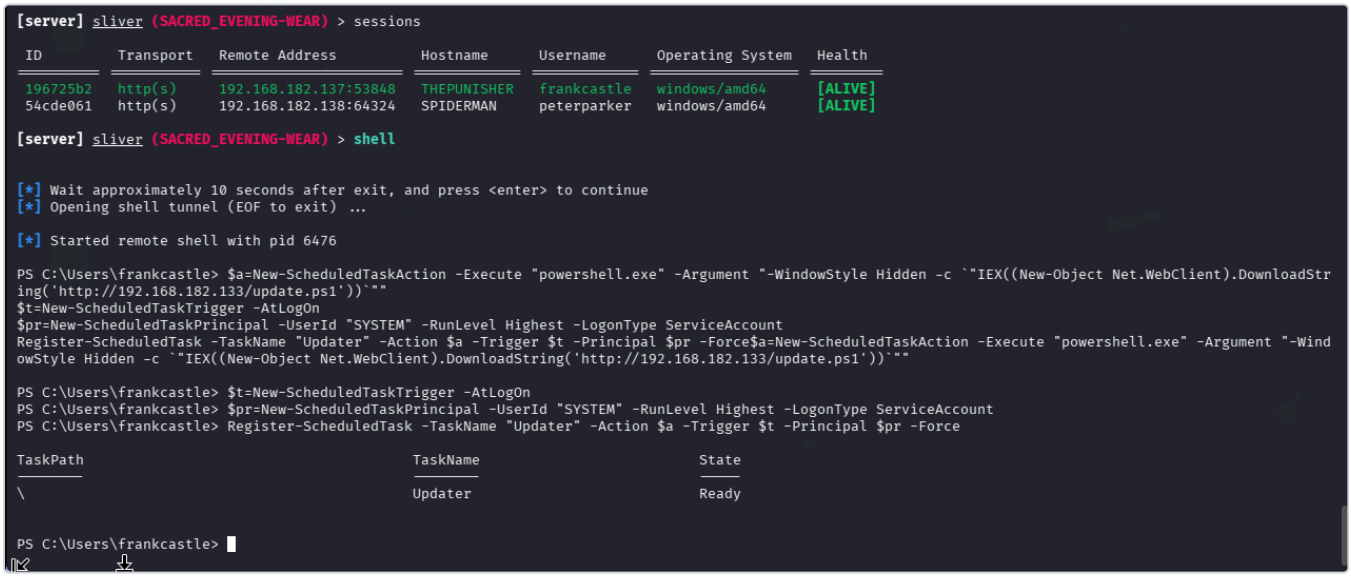
SharpKatz run in memory through Sliver on THEPUNISHER. This is the LSASS access caught by Sysmon EID 10 in detection 07, and missed by the EDR for the reasons in section 6.4.

Step 11 — Credentials recovered (T1003.001)



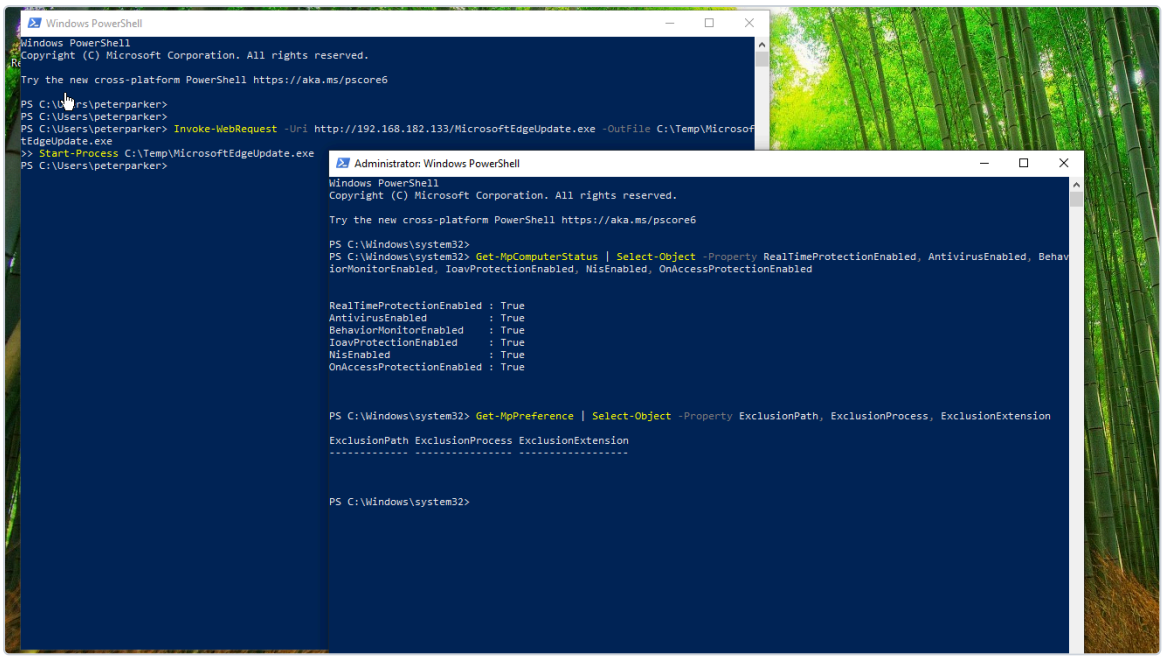
NTLM and SHA1 material, a cleartext `MARVEL\\administrator` credential, and the `THEPUNISHER$` machine account.

Step 12 — Persistence, scheduled task (T1053.005)



`Register-ScheduledTask -TaskName "Updater"` running `powershell.exe` as SYSTEM at logon. Its `4698` creation event is detection 08.

Step 13 — Defender enabled throughout (control state)



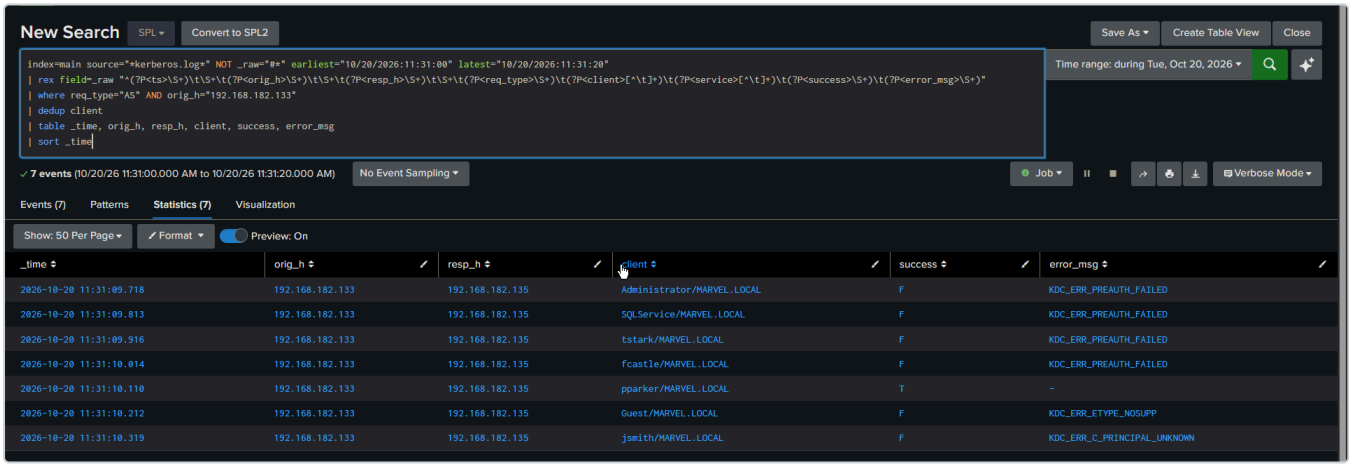
`Get-MpComputerStatus` shows real-time protection and all engines on, and `Get-MpPreference` shows no exclusion paths. A change from v2, which kept `C:\\Temp` excluded so delivery was reproducible on camera.

5. Detection coverage

Phase	Technique	Detected by
Password spray	T1110.003	Windows `4771`, Zeek `kerberos.log`
Kerberoasting	T1558.003	Windows `4769`, Suricata `krb5` and `weak_encryption`
Pass-the-Ticket	T1550.003	Windows `4768` TGT for `SQLService`, Zeek `kerberos.log`
Lateral movement	T1021.006	Zeek `conn.log` port 5985, Suricata `REQUEST_AUTH_UNRECOGNIZED`
LSASS access	T1003.001	Sysmon EID 10, THEPUNISHER only. Not by the EDR, see 6.4
Persistence	T1053.005	Windows `4698`, Wazuh `60228` at level 4
C2 beaconing	T1071.001	Zeek `ssl.log`, JA4 TLS fingerprinting

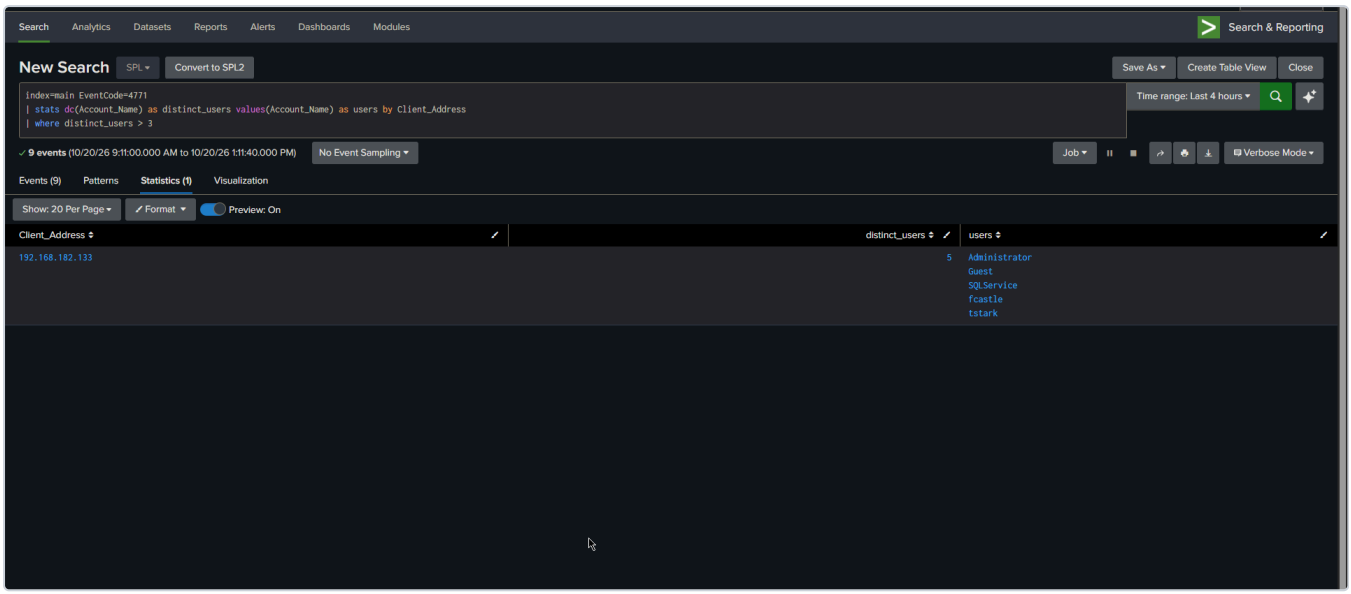
All times from the 20 October 2026 run. Zeek queries parse the raw TSV with `rex` to show log structure; in production the Zeek add-on extracts these fields, which is scoped into v4.

Detection 01 — Password spray, network view (T1110.003)



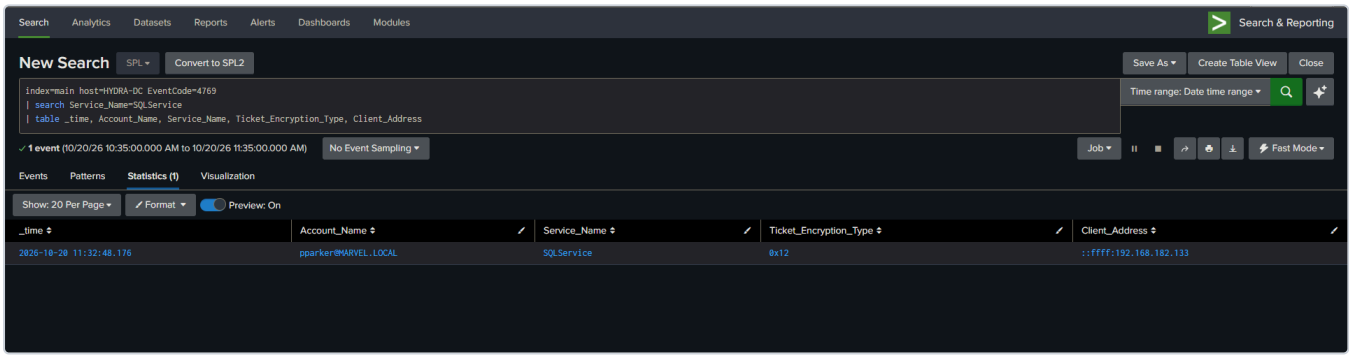
Zeek `kerberos.log`: seven accounts from `192.168.182.133` in 0.6 s. `pparker` succeeds. `jsmith` fails as unknown principal, which produces no Windows `4771` at all.

Detection 02 — Password spray, endpoint view (T1110.003)



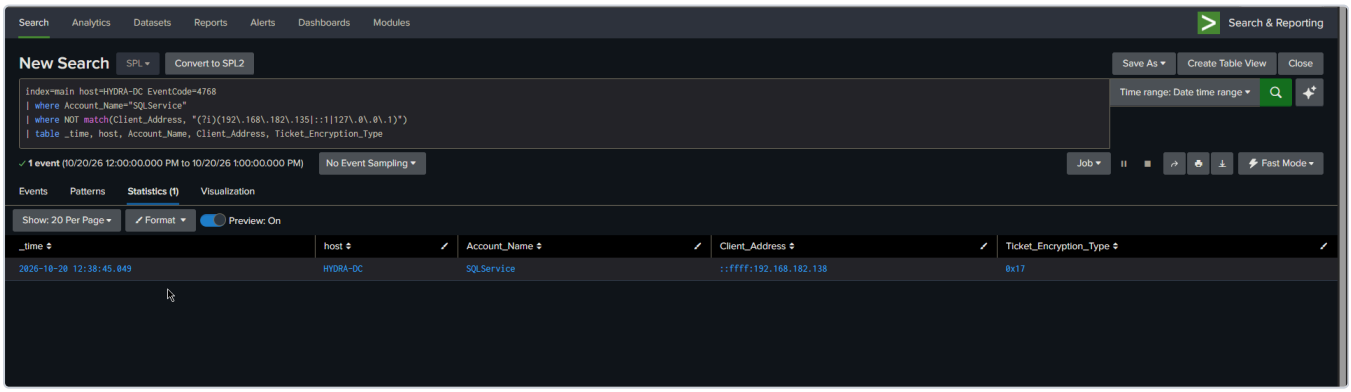
Windows `4771` correlation: single source, five accounts failing pre-authentication. Two fewer than the network layer saw.

Detection 03 — Kerberoasting, AES service ticket (T1558.003)



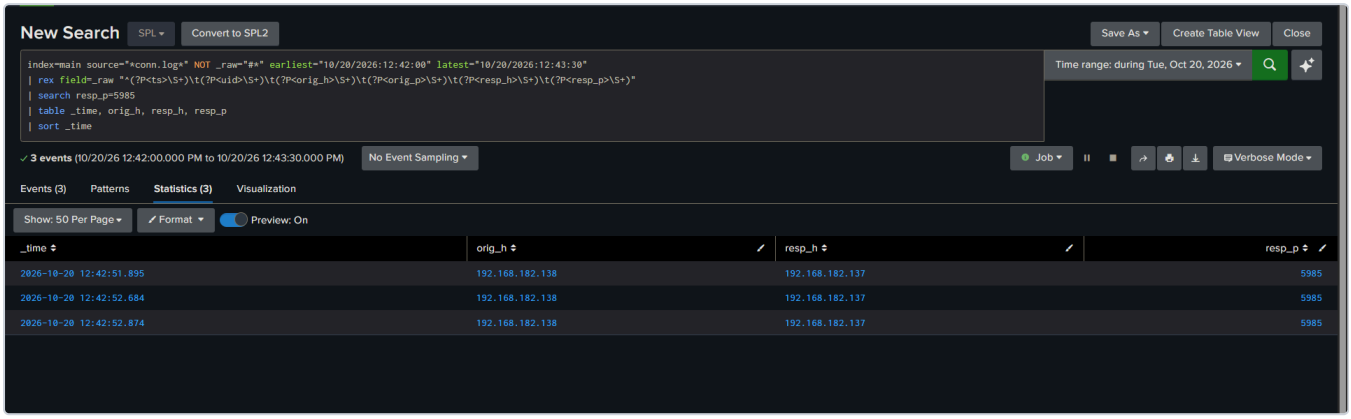
`4769` for `SQLService`, `Ticket_Encryption_Type=0x12` (AES-256), requested by `pparker`. The shipped RC4 rule keys on `0x17` and stayed silent.

Detection 04 — Pass-the-Ticket (T1550.003)



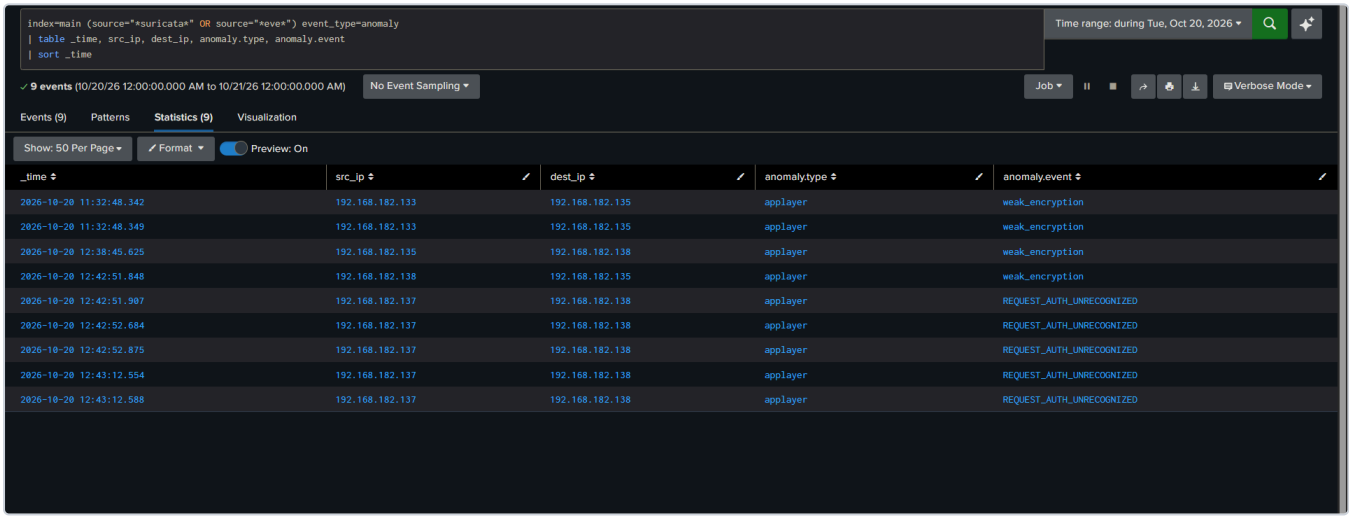
`4768` TGT requested *for* `SQLService` from SPIDERMAN at 12:38, RC4. `4768` rather than `4769` is the tell: a roast requests a service ticket, this requests a ticket-granting ticket in the account's own name, which means the credential is already held.

Detection 05 — Lateral movement over WinRM (T1021.006)



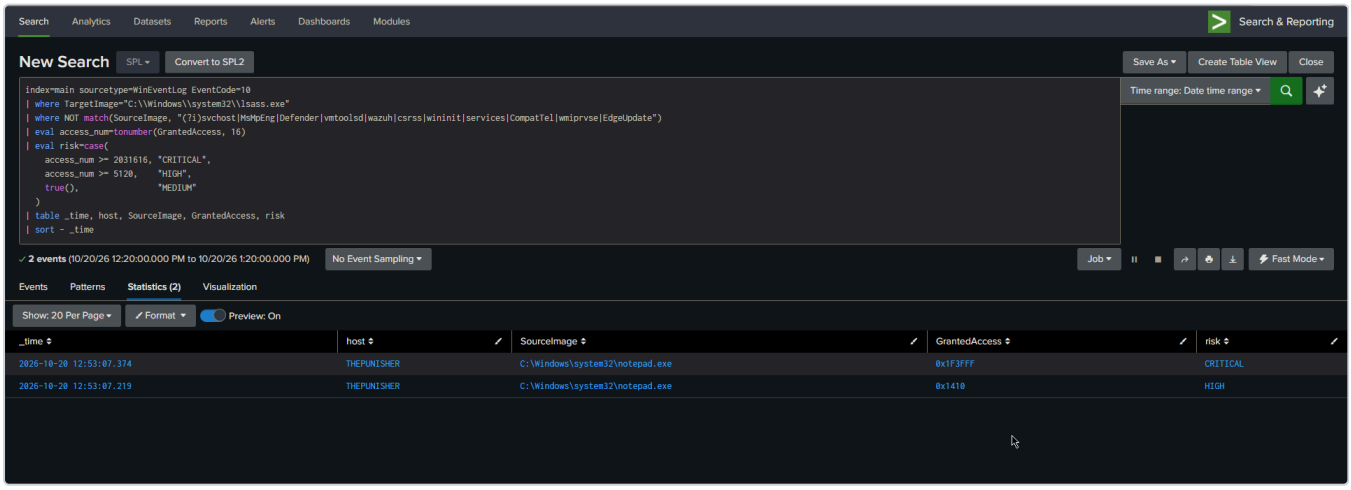
Zeek `conn.log`: three connections `192.168.182.138` to `192.168.182.137` on port 5985. Workstation to workstation, which has no legitimate reason in an estate with no jump host.

Detection 06 — Suricata application-layer anomalies



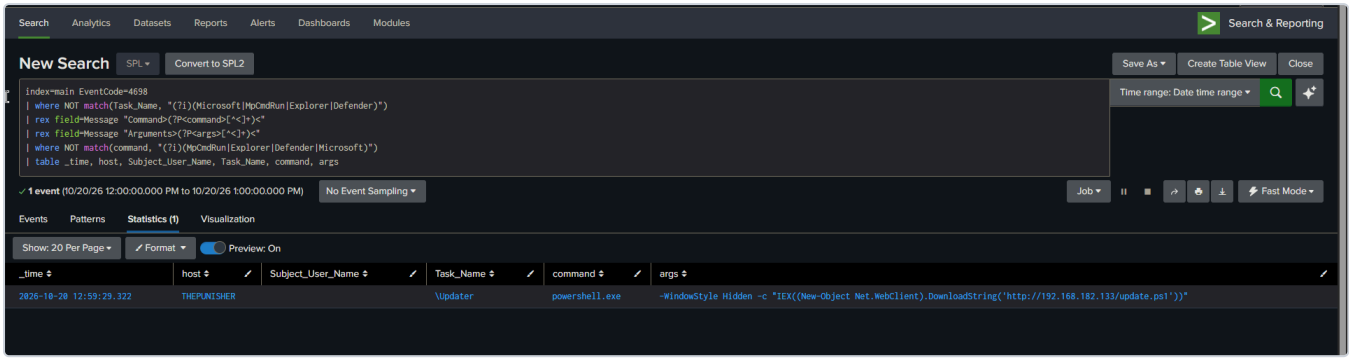
`weak_encryption` on the Kerberos exchanges, and `REQUEST_AUTH_UNRECOGNIZED` during the 12:42 to 12:43 WinRM pivot.

Detection 07 — LSASS access (T1003.001)



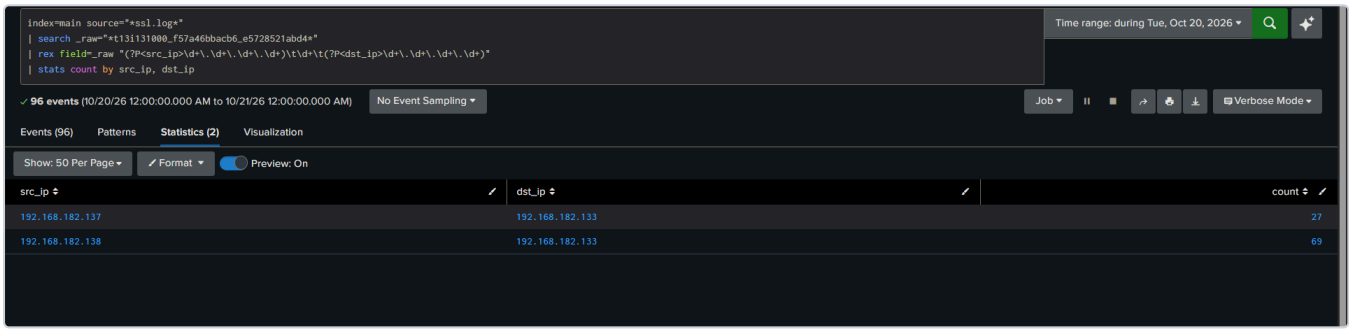
Sysmon EID 10 on THEPUNISHER: `notepad.exe` accessing `lsass.exe` with `GrantedAccess 0x1F3FFF`. The process name is a disguise; SharpKatz ran in memory inside a host process. The query excludes `MsMpEng.exe` so Defender's routine access is filtered out.

Detection 08 — Persistence, scheduled task (T1053.005)



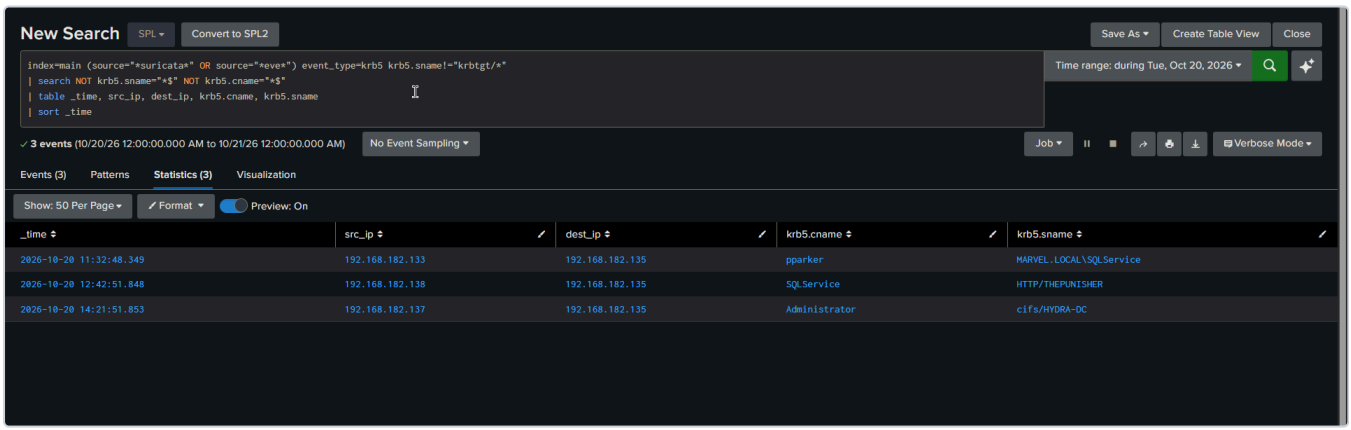
`4698` on THEPUNISHER: task `\\Updater`, `powershell.exe -WindowStyle Hidden -c IEX(...)`. The download cradle is in the event body, which the EDR rule does not read.

Detection 09 — C2 beaconing, JA4 fingerprint (T1071.001)



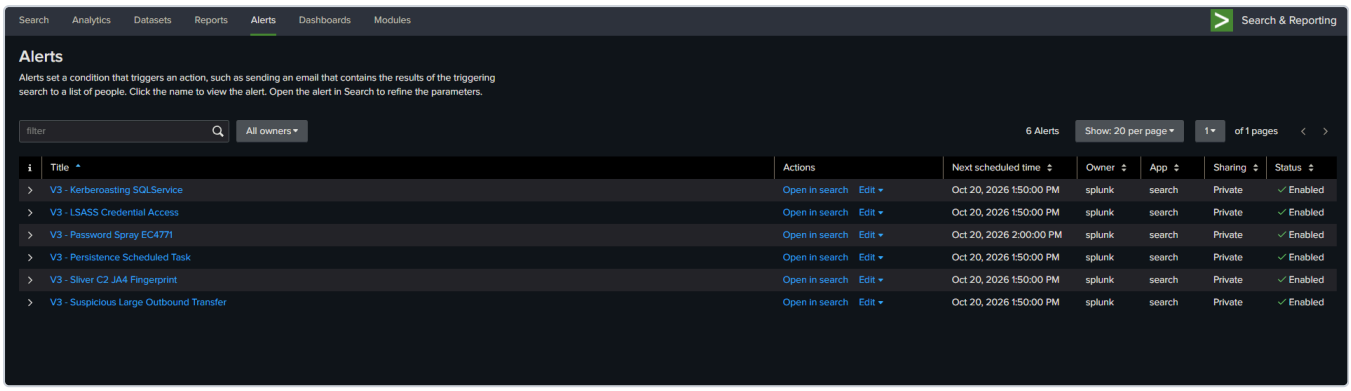
Zeek `ssl.log` filtered on the Sliver JA4 `t13i131000_f57a46bbacb6_e5728521abd4`: 69 beacons from SPIDERMAN and 27 from THEPUNISHER. One fingerprint links two compromises that share no filename or hash.

Detection 10 — Kerberoasting by signature, not by address (T1558.003)



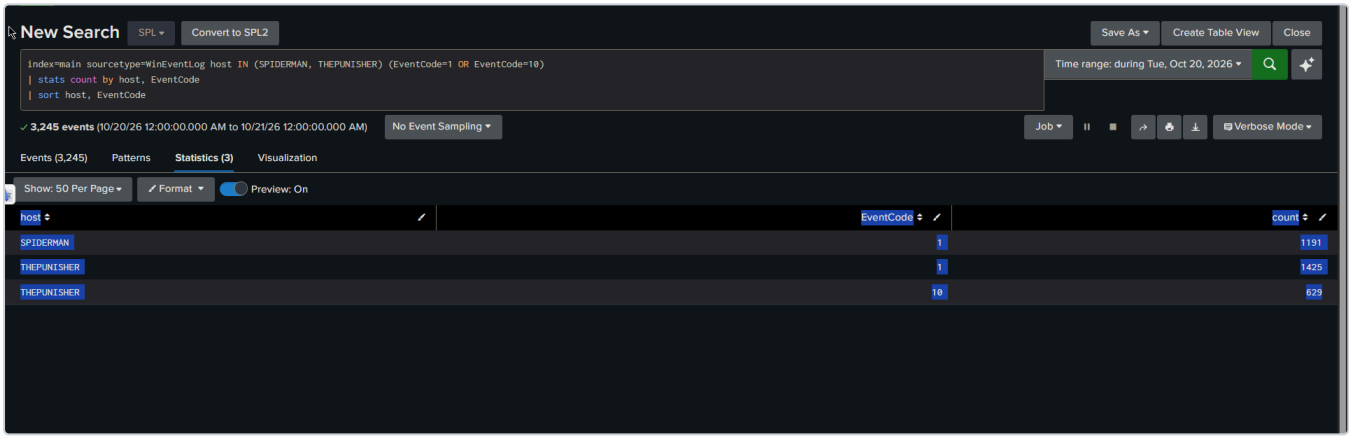
Suricata `krb5` filtered to service tickets whose SPN is neither `krbtgt` nor a machine account, so the filter keys on the pattern rather than the attacker's address. Three tickets surface the whole credential-abuse chain: the roast, the cracked account being reused against `HTTP/THEPUNISHER`, and `Administrator` against `cifs/HYDRA-DC`.

Detection 11 — Detection content shipped as alerts



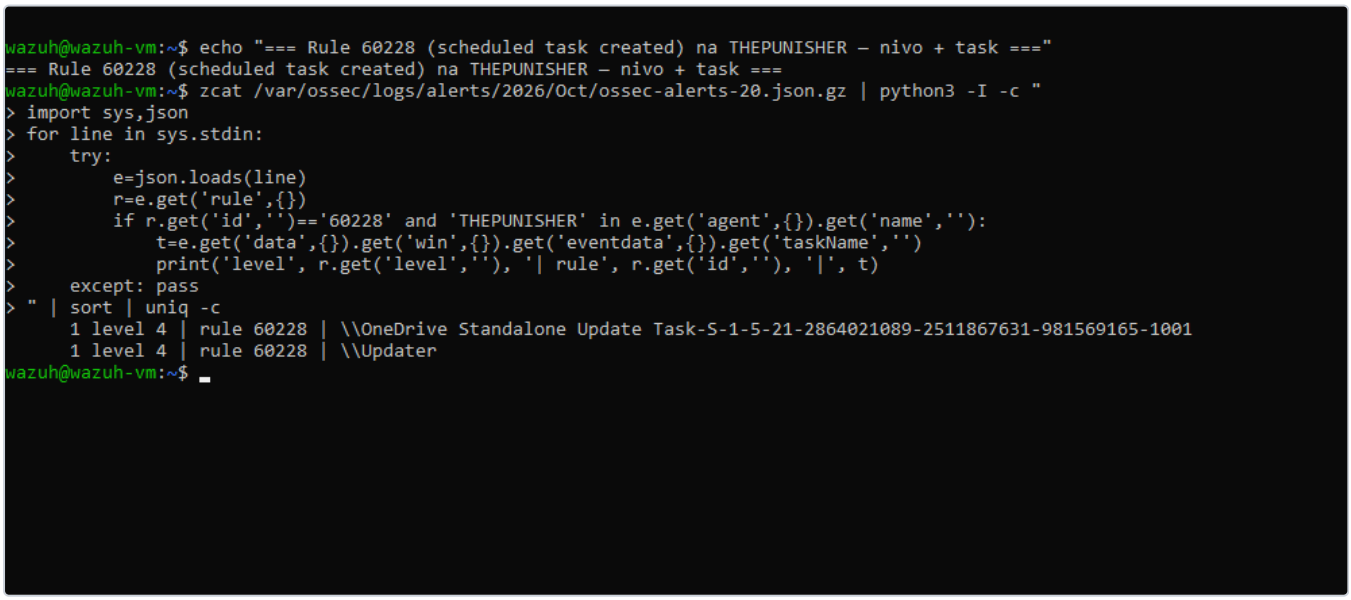
Six v3 saved alerts, all enabled. These fire on schedule because v3 runs an Enterprise licence; on v2's Free licence the same searches run only on demand.

Detection 12 — The SPIDERMAN blind spot, proven



SPIDERMAN forwarded 1,191 Sysmon process-create events and zero ProcessAccess events, while THEPUNISHER logged 629. `sysmon64 -c` on the host shows `ProcessAccess onmatch: include` with no rule lines under it, so the include list matches nothing. The same dump on that host would leave no trace.

Detection 13 — The Wazuh scheduled-task rule gap (T1053.005)



Rule `60228` fired at level 4 for the malicious `\\Updater` task and for a benign `\\OneDrive Standalone Update Task` . Same rule, same severity. The hidden PowerShell cradle inside `\\Updater` is in the event body and is not scored.

6. Findings

6.1 Zeek recorded a spray attempt Windows did not

Zeek `kerberos.log`, 11:31:09.718 to 11:31:10.319, source `192.168.182.133`:

Outcome	Accounts
Succeeded	`pparker`
Pre-authentication failed	`Administrator`, `SQLService`, `tstark`, `fcastle`
Unsupported encryption type	`Guest`
Unknown principal	`jsmith`

Windows logged `4771` for five of these. The `jsmith` attempt produced none, because Kerberos emits no `4771` for an account that does not exist. The Splunk alert reported five accounts; seven were tried.

The shipped rule narrows it further by filtering `Failure_Code="0x18"`, wrong password. Counting every `4771` failure code in this lab over 30 days:

Code	Meaning	Count
`0x18`	wrong password	67
`0xE`	encryption type not supported	7
`0x25`	clock skew	2

The seven `0xE` events are spray attempts that do produce a `4771`, and the rule drops them. So the endpoint layer undercounts twice: once for accounts that generate no event, and again for accounts that fail on something other than the password.

6.2 One JA4 fingerprint linked both endpoints

Both endpoints beaconed to the attacker over TLS with the same Sliver JA4 client fingerprint `t13i131000_f57a46bbacb6_e5728521abd4`: 69 sessions from SPIDERMAN, 27 from THEPUNISHER. The two implants were delivered at different times by different routes and share no filename or hash. The fingerprint is derived from how the TLS client is built, so renaming the binary does not change it.

It remains an indicator. It is tied to this Sliver build's TLS stack, and a different C2 or a changed client profile produces a different fingerprint. More durable than a hash, less durable than behaviour.

6.3 Wazuh scores task creation, not task content

The `\Updater` persistence task fired rule `60228` at level 4. A benign `\OneDrive Standalone Update Task` on the same host fired the same rule at the same level. The hidden PowerShell download cradle is in the `taskContent` field of the event, and nothing reads it.

Level 4 is below where an analyst working a severity-sorted queue looks. The event was received, parsed and alerted on, and the output was still useless.

6.4 Wazuh rule 92900 fired 32 times with 0 true positives

`92900` is Wazuh's built-in LSASS-access rule, level 12. Across all archived alert files:

Source	Count	Masks

`MsMpEng.exe` (Defender)	29	`0x101000`, `0x1010`
`svchost.exe`	2	`0x101000`
`MRT.exe`	1	`0x101000`

The SharpKatz read of LSASS is not among them. Reading the rule explains why:

```
<rule id="92900" level="12">
  <if_group>sysmon_event_10</if_group>
  <field name="win.eventdata.targetImage">(?i)lsass\.exe</field>
  <field name="win.eventdata.grantedAccess">(?i)(0x1010|0x40)</field>
```

`0x1010` is what Defender uses. `0x1fffff` and `0x1438` are not matched, and SharpKatz read LSASS at `0x1F3FFF`. The severity is inverted: wrong every time on the benign mask, silent on the aggressive one. Confirmed by trigger, a PowerShell `OpenProcess(lsass, PROCESS_ALL_ACCESS)` produced a Sysmon event and no Wazuh alert.

6.5 The EDR receives Sysmon from one of three agents

Agent	Alerts	From Sysmon	EID 10
HYDRA-DC	51,725	0	0
SPIDERMAN	22,782	0	0
THEPUNISHER	18,591	376	32

Only THEPUNISHER carries a `Microsoft-Windows-Sysmon/Operational` block in its `ossec.conf`. The DC case is the sharpest: Sysmon is installed and running there and its own log holds 43,683 events that the agent never reads. The agent reports healthy and busy while the EDR sees none of it.

The `4698` task rules are unaffected, because the Security channel is read everywhere.

6.6 The canonical Kerberoasting detection cannot fire on this domain

The RC4 rule is the one every guide teaches. Counted over 30 days of this lab's `4769` traffic:

Encryption type	Tickets
`0x12` (AES-256)	1,649
`0x17` (RC4)	0

Not one RC4 ticket. The rule is scheduled, it appears in the coverage table, and it has never had anything to match. The v3 roast was issued as `0x12` and walked past it.

This is the same failure as 6.4 reached from the other direction: there a rule could not load, here it loads and the environment never produces its input. Both report as coverage. The repo ships an etype-agnostic companion keyed on the SPN owner, which is noisier and needs a per-environment allowlist. On an AES domain there is no cheap high-confidence Kerberoasting signal, and that is the honest trade.

6.7 A rule tuned against the baseline rather than against the attack

A first draft of the LSASS replacement matched `0x1010` and `0x1410` alongside the full-access masks. Measured against 935 `ProcessAccess` reads of `lsass.exe` sampled from THEPUNISHER's own Sysmon log, it would have alerted 90 times:

Count	Mask	Source

79	`0x1410`	`WmiPrvSE.exe`
4	`0x1410`	`vmtoolsd.exe`
3	`0x1410`	`wazuh-agent.exe`
2	`0x1fffff`	`wazuh-agent.exe`, the EDR agent itself
1	`0x1410`	`taskmgr.exe`
1	`0x1f3fff`	`powershell.exe`, the only one worth triage

89 of 90 are ordinary platform behaviour. Restricting to full and MiniDumpWriteDump masks and excluding the sources measured above brings the same sample to one alert. A rule that is wrong 98.9 per cent of the time is not a detection, and the only way to know that in advance is to count.

7. Detection engineering

What ships

Artefact	Contents
Sigma	13 vendor-agnostic rules, 10 host and 3 network
Splunk app	13 saved searches, 11 on raw sourcetypes and 2 on CIM data models
Wazuh	5 custom rules: LSASS detection, suppression of the built-in false positive, and task-content scoring
Sysmon config	including the `NetworkConnect` gap fix from v2

The two failures worth documenting

A rule that could not load

The original LSASS rule was deployed against `<if_group>sysmon_event10</if_group>`. The real group is `sysmon_event_10`, one underscore away. A `grep` over the ruleset returns nothing for the misspelling, and no alert in any archive has ever carried it. The rule alerted zero times, ever.

Its silence was first read as "no attack has been run since it was added", which was also true, so two causes produced the same zero and the harmless one hid the real one. Wazuh had been reporting the fault on every start:

```
wazuh-analysisd: WARNING: (7610): Group 'sysmon_event10' was not found.
Invalid 'if_group'. Rule '100200' will be ignored.
```

A replacement that also could not fire

The first replacement bound to `<if_sid>92900</if_sid>`, reasoning that a concrete parent SID is more stable than a group name. That could not fire either: `92900` filters `grantedAccess` to `0x1010` or `0x40`, so a child requiring a full-access mask is mutually exclusive with its own parent. Caught by running it rather than by reading it.

Proof that the fix works

A controlled before and after fell out of the test run. The same trigger, from the same host, with the same mask, six minutes apart, with only the rule's parent changed between them:

Time (UTC)	Parent	Alert
11:36:58	`<if_sid>92900</if_sid>`	none
11:42:38	`<if_group>sysmon_event_10</if_group>`	level 12

Sysmon recorded both events. The full verification log is Appendix A.

Where the Splunk pack does and does not work

Component	State in this lab
`Endpoint.Processes` data model	populated; the LOLBin detection works
`Network_Traffic.All_Traffic` data model	**empty**; the C2 beacon detection cannot return a row

Data model acceleration	**off** on both, so `tstats` here is a raw scan
The other eleven searches	work, because they filter on `source=` rather than `sourcetype=`

The cause is the forwarder. Every Windows channel arrives as a bare `WinEventLog` sourcetype with the channel name only in `source`:

```
sourcetype = WinEventLog
source     = WinEventLog:Microsoft-Windows-Sysmon/Operational
```

The Sysmon add-on keys its CIM field aliases and eventtypes on the per-channel sourcetype. Under a bare `WinEventLog` none of it applies, so 444 EID 3 events in seven days reach the index with no `app`, `src`, `dest` or `dest_port`, and nothing tags them into the model.

That the other eleven rules survive this was not a deliberate choice at the time. It is the accident of having written them against `source=`. The fix is a forwarder change rather than a rule change, and it is scoped into v4.

> The affected search ships unscheduled with the reason written into its description. An empty data model and a broken rule look identical from the alert list, which is precisely why it has to be stated rather than quietly fixed.

8. Alert triage

Detections are half the job. The other half is what an analyst does when one fires, and that is the part a SOC role is measured on. Seven case files cover the chain phase by phase in one format: the alert, the questions asked, the queries run, the verdict, and the tuning and hardening that follows. They chain in the order the attack ran.

Case	Phase	Verdict
CASE-01	Password spray	True positive. Seven accounts in 0.6 s from a non-domain host, one succeeded, used 98 seconds later.
CASE-02	Kerberoasting	True positive. Domain Admin credential cracked offline; the canonical RC4 rule never fired.
CASE-03	Pass-the-Ticket	True positive. `4768` rather than `4769` separates the theft from the use.
CASE-04	Lateral movement	True positive. The firewall mesh chose the route; WinRM was the one open lane.
CASE-05	LSASS access	True positive on one host, undetectable on the other. Four layers of absent coverage.
CASE-06	Persistence	True positive delivered at level 4, below where anyone looks.
CASE-07	C2 beaconing	True positive. Two endpoints linked by one fingerprint across the whole run.

Three of them are about detections that did not work, which is why they are written out rather than summarised. CASE-02 covers a rule that cannot fire on an AES domain, CASE-05 covers four separate reasons an EDR stayed silent on a credential dump, and CASE-06 covers an alert that arrived correctly and still told the analyst nothing.

Full text in `v3/investigations/` in the repository.

9. Known gaps carried into v4

The hardening baseline is partial and inconsistent

That is the starting point rather than an oversight. v4 rebuilds it to a CIS-aligned standard: consistent SMB signing, LSA Protection, WDigest and LLMNR off, and a GPO firewall baseline with a domain-isolation gate, with WinRM east-west left open as the monitored pivot. The chain is then re-run for a before and after coverage matrix.

Account lockout is enforced and the spray never reached it

One attempt per account per window is how spraying evades lockout. The weaker control here is the seven-character minimum password length, which v4 raises.

The service account is a human-managed Domain Admin with an SPN

The roast cracks straight to Domain Admin rather than to a low-privilege service account, and the password was also stored in the account's AD `Description` field, which is a second exposure needing no cracking. v4 moves it to a gMSA and clears the description.

The Wazuh gaps are fixed at the rule layer, not the telemetry layer

The rules are deployed and verified, so `\Updater` now scores level 12 and a full-access LSASS handle alerts at all. The ingestion gap is not fixed: the LSASS rule has telemetry on one of three hosts. A correct rule over missing data is still no coverage.

The forwarder sends a bare sourcetype

Measured, not planned. Per-channel sourcetypes with `renderXml = 1`, then acceleration enabled, then both `tstats` detections re-run with a before and after. "The data model is empty" and "the rule is wrong" look identical from the alert list, so the fix has to be demonstrated rather than assumed.

Zeek is parsed positionally

v3 queries parse the raw TSV with `rex`, which shows log structure but breaks on an empty or shifted column. v4 switches Zeek to JSON and ingests through the Splunk add-on, so fields are extracted by the TA.

Appendix A — Rule verification log

24 October 2026. Every rule deployed and fired by a targeted trigger rather than assumed to work. Times UTC; the Windows hosts run UTC+2.

Rule	Trigger	Result
—	`wazuh-analysisd -t`	clean; the previous version raised `(7610)` and discarded its rule
`100200`	PowerShell `OpenProcess(lsass, PROCESS_ALL_ACCESS)`	level 12 at 11:42:39
`100201`	20 platform `ProcessAccess` reads forced through the chain	0 alerts
`100205`	`MsMpEng.exe` reading LSASS at `0x1010`	0 alerts, where it was level 12 thirty-two times out of thirty-two
`100210`	task with `IEX` and `DownloadString`	level 12 at 11:48:57
`100211`	task with `-nop -w hidden -enc`	level 10 at 11:49:04
negative	task running `defrag.exe C: /O`	stays `60228`, level 4
negative	task running `powershell.exe -ExecutionPolicy Bypass -File ...`	stays `60228`, level 4

The second negative test is the one that counts. A scheduled task that legitimately runs PowerShell with `-ExecutionPolicy Bypass` is the realistic false positive for these rules, and neither fired on it. Writing a rule that alerts is easy; writing one that stays quiet when it should is the part worth testing.

`100201` has no direct trigger, because no platform process in this lab opens LSASS with a full-access mask. It was tested by temporarily widening `100200` to include `0x1000`, the mask `svchost.exe` and `MsMpEng.exe` use constantly, so ordinary platform reads had to pass through the suppression chain. Twenty such events produced zero alerts, and the mask was then reverted.

Appendix B — Measurement data

LSASS access masks present in the index

Sysmon EID 10 against `lsass.exe`, counted over seven days:

Mask	Count	In the shipped rule
`0x1000`	1,498	no
`0x1410`	177	no, this is the 79-event WmiPrvSE noise plus others
`0x1400`	107	no
`0x101000`	40	no
`0x1010`	25	no
`0x1418`	4	no
`0x1FFFFF`	8	**yes**
`0x1F3FFF`	2	**yes**, this is the attack

The shipped mask list is `0x1438`, `0x143a`, `0x1f0fff`, `0x1f1fff`, `0x1f2fff`, `0x1f3fff` and `0x1fffff`, identical across the Sigma rule, the Splunk search and the Wazuh rule.

Kerberos ticket encryption types, 30 days

Type	Count
`0x12` (AES-256)	1,649
`0x17` (RC4)	0

Kerberos pre-authentication failure codes, 30 days

Code	Meaning	Count
`0x18`	wrong password	67
`0xE`	encryption type not supported	7
`0x25`	clock skew	2

Sysmon event IDs present on SPIDERMAN

Grouping 5,000 recent events by ID returns eleven distinct values: 1, 2, 3, 4, 5, 8, 11, 12, 13, 22 and 255. EID 10 does not appear.

Splunk environment

Item	Value
Licence group	Enterprise, so scheduled searches and alerting are available
Sourcetypes in the index	`WinEventLog` 330,188 · `suricata` 139,759 · `zeek` 131,888 · `wazuh` 33,042
`Endpoint.Processes`	populated

`Network_Traffic.All_Traffic`	0 events
Acceleration	not enabled on either model

> Every figure in this appendix came from a query run against the live index or from reading the running configuration on the host, not from the attack run's own output. Where a number could not be confirmed that way, it is not here.